ChainWork

Decentralised Freelance Platform

Team LedgerLogic

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Decentralization: A Better Approach

Challenges in Centralized Freelance Platforms

- Intermediary Dependence
- Payment Delays
- High Fees
- Manual Disputes
- Limited Transparency

The Alternative

- Decentralized Freelancing Model
- Escrow-Based Payment System
- On-Chain Reputation System
- Trustless Transactions using Smart Contracts

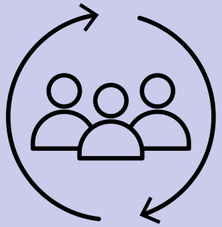
core features:

Smart Contracts

Automates hiring, payments, and job completion securely

Escrow Payment System

Funds are locked safely until work is completed

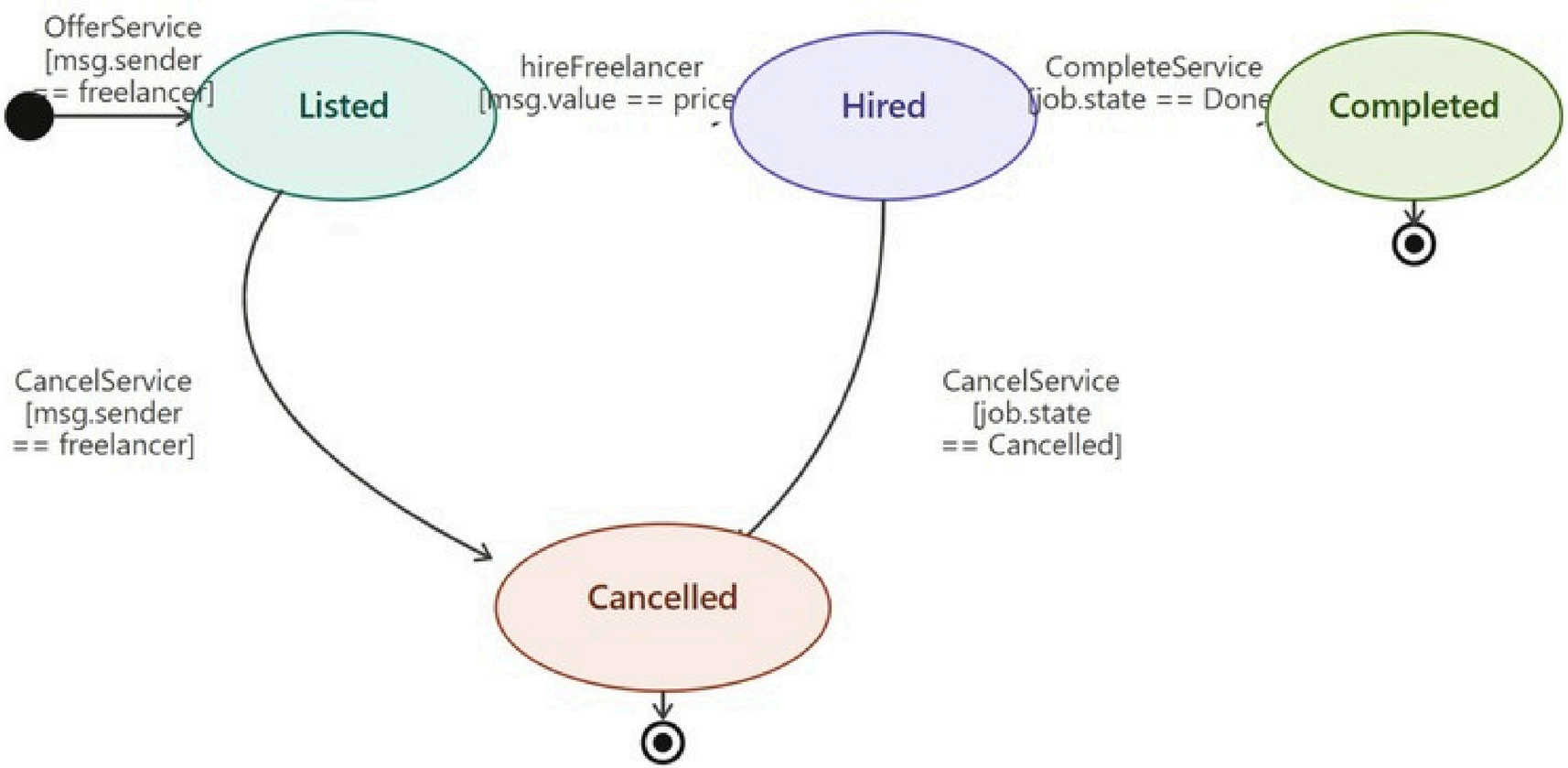


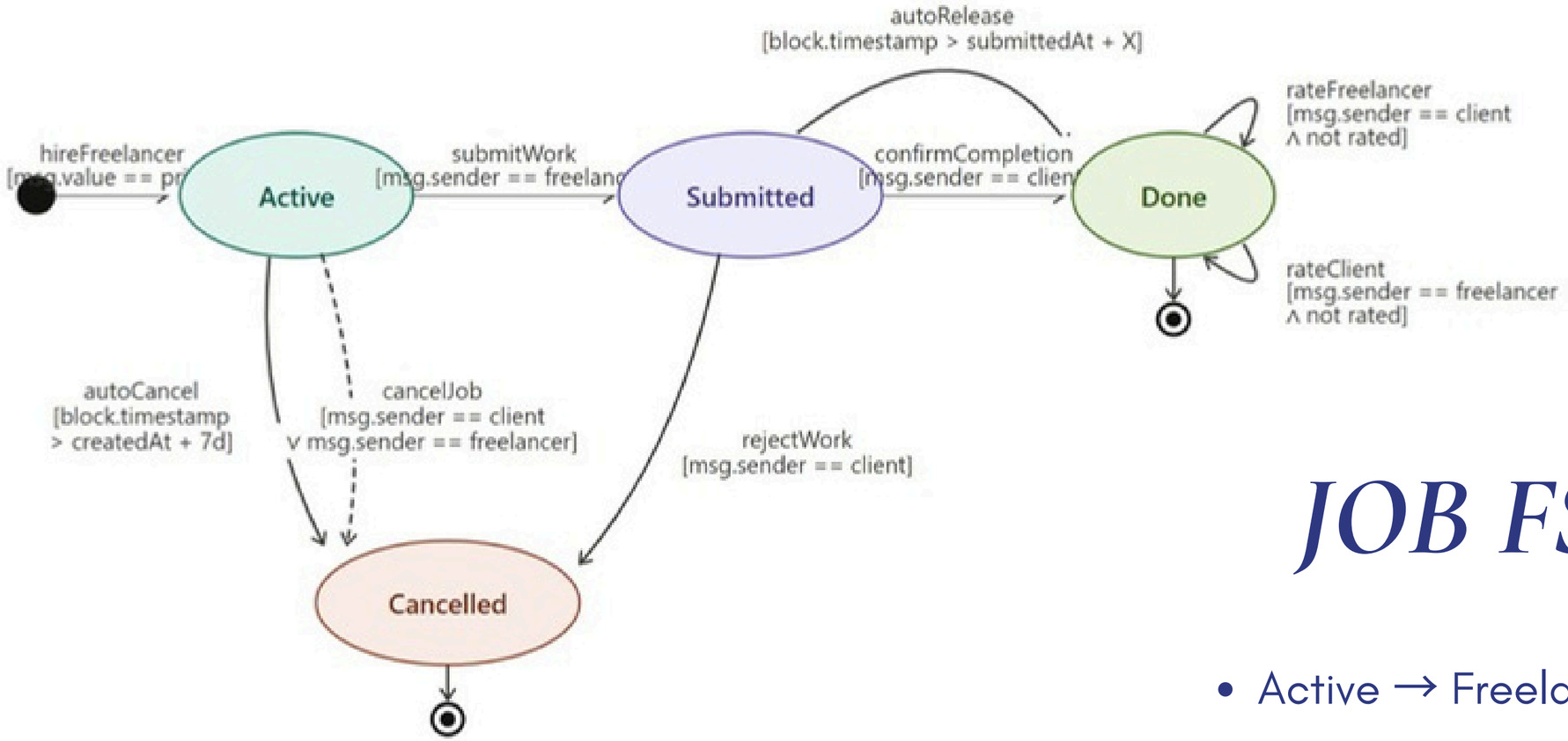
Reputation System

Freelancers are rated (1–5) and scores stored on-chain

Service FSM

- Listed → Service available for hiring
- Hired → Client selects service & locks payment
- Completed → Work finished successfully
- Cancelled → Service stopped before completion





JOB FSM

- Active → Freelancer is working
- Submitted → Work delivered
- Done → Client approves & payment released
- Cancelled → Job terminated

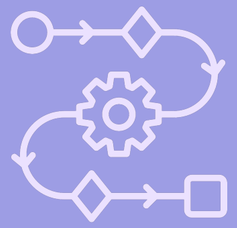
On Chain and Off Chain Storage

On Chain:

- Wallet addresses (freelancer, client)
- ETH amounts and escrow balances
- Job status (Active / Submitted / Done / Cancelled)
- Deadline and timestamps
- File hashes — metadataCid, workCid, jobDescription
- Reputation scores and weights
- Feedback tokens and stake balance

Off Chain:

- Service title, description, images, tags
- Full job brief text
- Deliverable files submitted by the freelancer



Client-Side Workflow

1. Stake Initialization

depositStake()

- Client invokes depositStake() to register participation
- Requires a minimum stake of 0.05 ETH
- Enforces Sybil resistance via economic barrier
- Mandatory prerequisite for accessing feedback functions
- Stake is non-slashable and remains locked as commitment
- Acts as identity verification through capital bonding

2. Job Creation & Escrow Locking

hireFreelancer(serviceId, deadline, jobDescription):

- Client invokes hireFreelancer(...) to instantiate a job contract
- Must transfer exact service price (msg.value == priceWei)
- Enforces constraint: caller ≠ service owner

Parameters:

- serviceId → reference to on-chain service struct
- deadline → bounded within [1, 365] days
- jobDescription → fixed-size bytes32 metadata field

State Transition:

- Job state initialized to Active
- Funds transferred to contract and held in escrow (contract balance)

3. Job Finalization

confirmCompletion(jobId):

- Client invokes confirmCompletion(jobId) to finalize job

Execution Semantics::

- Validates caller authorization (only client)
- Applies Checks-Effects-Interactions (CEI) patternState updated before external transfer
- Transfers escrowed ETH to freelancer

Post-State::

- Job state transitions to Completed
- Submitted data becomes immutable (on-chain persistence)
- Mints dual feedback tokens for both participants

4.Cancellation Logic & Edge Cases

cancelJob(jobId):

- Invoked to terminate job prior to completion

State-Based Access Control:

Active State:

- Callable by:

1. Client
2. Freelancer
3. Any address after deadline expiry

Submitted State:

- Before deadline → client-only access
- After deadline → permissionless (any caller)

Execution Outcome:

- Escrow funds refunded to client
- Job state transitions to Cancelled
- No state mutation in reputation mappings

5. Feedback Submission

submitFeedback(tokenId, score):

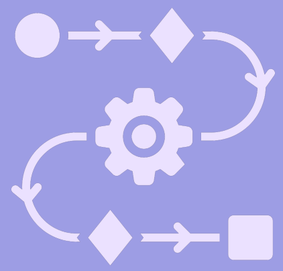
- Client invokes submitFeedback(...) using assigned tokenId
- Score constrained to uint8 range [1-5]

Preconditions:

- Token must be valid and unused
- Associated job must be Completed

Effect:

- Records rating in reputation aggregation logic
- Contributes to weighted scoring mechanism



Freelancer Workflow

1. Stake & Service Listing

Stake Deposit (depositStake)

- Minimum 0.05 ETH required
- Acts as Sybil resistance (prevents fake users)
- Required to participate in feedback/reputation system

Service Listing (offerService)

- Off-chain via IPFS: description, portfolio, files
- On-chain: freelancer address, price, metadata CID
- Service becomes visible to clients (Listed state)

2.Hiring & Escrow

Client Hiring (hireFreelancer):

- Client selects service & sends exact payment
- Cannot hire own listing (security check)

Escrow Mechanism:

- Funds locked using Smart Contracts
- Job created with Active status
- Deadline set for completion

Benefit:

- Guarantees payment availability before work starts

3.Work Submission

Work Upload:

- Freelancer uploads work to IPFS
- Generates Content Identifier (CID)

On-Chain Record (submitWork):

- CID stored on blockchain
- Job status → Submitted
- Work becomes tamper-proof & verifiable

Client Interaction:

- Client reviews submitted work before approval

4.Payment Resolution

Successful Completion:

- Client calls confirmCompletion()
- Payment released securely

Auto Release Mechanism:

- If no response → auto-release after 3 days

Cancellation Rules:

- Job cancelled → refund to client
- No reputation impact

Security:

- Uses escrow + deadlines to prevent fraud

Reputation System

Theoretical Features

1. Stake-Based Identity System (Sybil Resistance)

Implementation:

- The contract requires a MIN_STAKE of 0.05 ETH (~₹11,500) to be deposited before any rating can be submitted

Justification:

- This creates an Identity Tax
- In a decentralized world, creating accounts is free
- By adding a financial barrier, we ensure every review is backed by "skin in the game"
- Makes it prohibitively expensive to create fake accounts for "bad-mouthing" or "ballot-stuffing"

2. Mutual Accountability (The Dual-Token Model)

Implementation:

- Reputation tokens are permission-based
- Exactly two unique tokens are issued only upon job completion (confirmCompletion)

Justification:

- Provides On-Chain Proof of Service
- You cannot leave a review unless a transaction was successfully completed
- Ensures a balanced marketplace where both parties are held to professional standards

3. The "Blind Ballot" (Anti-Retaliation Logic)

Implementation:

- A "Holding Vault" mechanism
- Scores are submitted but not applied to the public reputation until:
  - both parties have voted, OR
  - a 7-day timeout occurs

Justification:

- Solves the "Hostage Rating" problem
- Users can be 100% honest without fearing a "revenge" 1-star rating
- Scores are revealed and applied simultaneously

4. Fixed-Point Precision Scaling (UX Control)

Implementation:

- The contract multiplies the final average by 100 before division

Justification:

- Solidity does not handle decimals
- This scaling allows the UI to display a clean 1.00 to 5.00 rating
- Provides high-resolution trust data (e.g., 4.85) without rounding errors

5. Future Scope: Feedback Incentives (The Discount Model)

Implementation:

- While not currently in the core escrow code, the system is designed to trigger a Service Fee Rebate on the user's next job

Justification:

- By scaling a future discount based on the review's "Weight", we turn reputation into a financial asset
- Creates a Loyalty Loop, rewarding users for high-quality, fast, and significant feedback

Formula Flow
(The Mathematical Engine)

To arrive at the final rating, the system follows a logical progression from individual job metrics to a global trust score.

Step 1: Calculate Individual Review Weight (w)

- The "gravity" of a single review depends on:
  - the job's value
  - the reviewer's responsiveness

$$amountWeight = \frac{\max(\text{JobAmount}, 1\ \text{Finney})}{1\ \text{Finney}}$$

$$speedWeight = \max(7 - \text{DaysToReview}, 1)$$

$$w = amountWeight \times speedWeight$$

Step 2: Update the Reputation Ingredients

Weighted Score Sum (WSS): The total "points" earned

- The contract does not store the average
- It stores the "Weighted Sums" to maintain accuracy over time

Step 2: Update the Reputation Ingredients

Weighted Score Sum (WSS): The total "points" earned

$$WSS_{new} = WSS_{old} + (\text{Score} \times w)$$

Weight Sum (WS): The total "power" of all reviews combined

$$WS_{new} = WS_{old} + w$$

Step 3: Calculate the Public Rating (R)

$$R = \frac{WSS \times 100}{WS}$$

Result Interpretation: An integer between 100 and 500.Example: A result of 485 translates to a 4.85-star rating on the platform UI.

Gas Optimization

- **Storage Efficiency**
  - Old Contract: Used uint256 for all variables (IDs, prices, timestamps), consuming a full 32-byte slot each and increasing storage costs.
  - New Contract: Applied Tight Struct Packing using smaller types (uint32 IDs, uint88 prices, uint64 timestamps), fitting multiple values into one 32-byte slot.
  - Result: Reduced Job storage from 7 slots to 3 slots, cutting user gas fees by ~60%.

- **Increment Logic**
  - Old Contract: Used jobCount = jobCount + 1;, adding extra overhead.
  - New Contract: Uses prefix increment ++jobCount;.
  - Result: Slight gas savings per transaction

Error Handling (Custom Errors)

- **Old Contract:** Used require(..., "long error string"), increasing on-chain storage and revert costs.
- **New Contract:** Switched to Custom Errors (error InsufficientStake()), using a 4-byte selector.
- **Result:** Lowered deployment gas and revert execution gas.

Security & Compiler Precision

- **Old Contract:** Used floating pragma ^0.8.20, allowing version inconsistency.
- **New Contract:** Pinned pragma to 0.8.28.
- **Result:** Exact tested compiler, latest patches, audit-ready.

Memory Management

- Old Contract: Re-read storage variables multiple times per function.
- New Contract: Used storage caching (Job storage j = jobs[jobId]).
- Result: Fewer SLOADs, lower gas costs.

Gas Optimization Comparison

Method	**old gas**	**New gas**	**Comparison**
hireFreelancer	151K	85K	↓ ~43.6%
submitFeedback	168K	117k	↓ ~30.3%
confirmCompletion	312k	257k	↓ ~17.5%
submitWork	95k	59k	↓ ~37.8%
offerService	113k	91k	↓ ~19.2%
cancelJob	63K	47k	↓ ~24.9%

Optimized Gas Report

Solidity and Network Configuration				
Solidity: 0.8.28	Optim: false	Runs: 200	viaIR: false	Block: 60,000,000 gas
Methods				
Contracts / Methods	Min	Max	Avg	# calls	usd (avg)
FreelanceEscrow					
autoRelease	–	–	258,097	9	–
cancelJob	46,743	49,499	47,833	16	–
clearWork	–	–	27,955	4	–
confirmCompletion	–	–	257,852	37	–
depositStake	28,153	45,253	44,846	42	–
finalizeReview	33,094	119,385	61,923	9	–
hireFreelancer	–	–	85,537	66	–
offerService	74,604	91,704	91,466	72	–
submitFeedback	41,301	205,814	117,221	39	–
submitWork	–	–	59,175	50	–
Deployments				% of limit	
FreelanceEscrow	–	–	3,784,268	6.3 %	–
Key					
○ Execution gas for this method does not include intrinsic gas overhead					
△ Cost was non-zero but below the precision setting for the currency display (see options)					
Toolchain: hardhat					

Unoptimized Gas Report

Solidity and Network Configuration				
Solidity: 0.8.28	Optim: false	Runs: 200	viaIR: false	Block: 60,000,000 gas
Methods				
Contracts / Methods	Min	Max	Avg	# calls	usd (avg)
FreelanceEscrow					
autoRelease	–	–	314,895	9	–
cancelJob	52,492	72,063	63,678	15	–
clearWork	–	–	29,760	4	–
confirmCompletion	–	–	312,733	29	–
depositStake	28,251	45,351	44,901	38	–
finalizeReview	33,135	127,274	64,580	9	–
hireFreelancer	–	–	151,684	46	–
offerService	96,390	113,490	113,147	50	–
submitFeedback	86,630	258,799	168,183	38	–
submitWork	–	–	95,074	37	–
Deployments				% of limit	
FreelanceEscrow	–	–	3,814,439	6.4 %	–
Key					
○ Execution gas for this method does not include intrinsic gas overhead					
△ Cost was non-zero but below the precision setting for the currency display (see options)					
Toolchain: hardhat					

Debugging tools

Slither:

- Static analysis on Solidity source code
- Detects:
  - Reentrancy vulnerabilities
  - Access control issues
  - Uninitialized variables
  - Gas inefficiencies
- Very fast → ideal for early-stage checks

Mythril:

- Uses symbolic execution on compiled bytecode
- Explores multiple execution paths
- Detects:
  - Reentrancy attacks
  - Integer overflow / underflow
  - Logical flaws in contract flow
- Provides deep runtime-level insights

Our contract showed no critical vulnerabilities in either tool.

Tech Stack

- Blockchain Layer: **Solidity**
- Development Framework: **Hardhat v2**
- Blockchain Interaction: **Ethers.js**
- Frontend Layer: **React.js**
- Off-Chain Storage: **IPFS** (Future Implementation)
- Wallet Integration: **MetaMask**
- Smart Contract Security: **OpenZeppelin**
- Programming Languages: **Solidity, JavaScript**

Improvements Made After Presentation

Discount Token:

- Implemented discount tokens as a proof of concept.
- Users (clients/freelancers) receive tokens for submitting feedback.
  - Token value depends on:
  - How quickly the review is submitted
- The size of the job/payment
- Currently, the tokens hold no monetary value, but could gain on-chain utility through a future slashing-based mechanism.

Email for Communication:

- Both client and freelancer now submit their email id's in their profile

Cancellation Fees:

- Added a cancellation_fee attribute to the Job struct.
- The client and freelancer mutually agree on the cancellation fee before the job begins.
- If the client cancels the job after work submission, the fee is paid to the freelancer.
- This prevents clients from misusing submitted work and cancelling the job without compensation, while maintaining fairness for both parties.

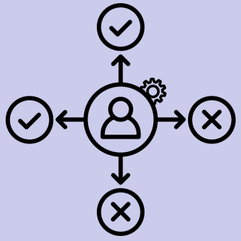
Future Improvements

Discount Token Incentives

Implemented as a proof of concept so far
Can be improved with slashing

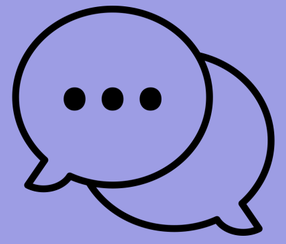
Fraud Detection & Slashing

Use pattern recognition to detect misuse and penalize malicious users



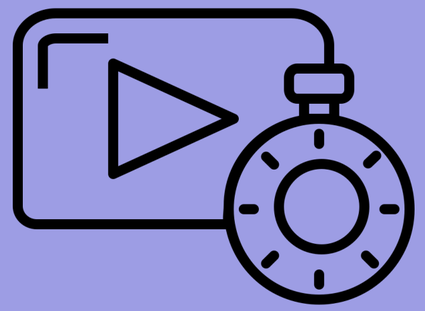
Decentralized Dispute Resolution

Resolve conflicts without a central authority



Decentralized Chat (Off-Chain Consensus)

Enable secure communication between client and freelancer for agreement before on-chain actions



Demo Video

- https://drive.google.com/file/d/1TpGnNTOWfoc1PsrbHJCbasG47uMAEqqE/view?usp=drive_link

- https://drive.google.com/file/d/1l-LEgkSEOIg9RbfLtAG0BlwCZYusNwd6/view?usp=drive_link

Thank You

LedgerLogic